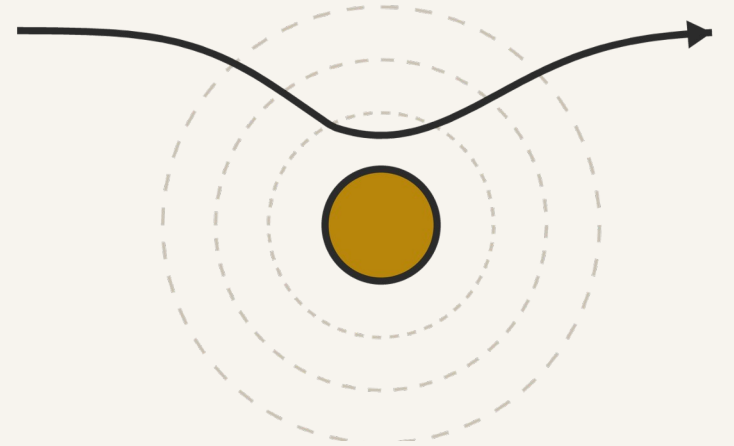


General relativity

Gravity is not a force. Here is what Einstein put in its place.

Advanced Higher Physics



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

LEARNING INTENTION

To state the equivalence principle and its consequences, to describe spacetime, world lines and geodesics, and to calculate the Schwarzschild radius of a mass.

By the end of this topic...

- 1 I can state the equivalence principle and describe its consequences.
- 2 I can describe spacetime, and draw and read world lines.
- 3 I can explain why a clock lower in a gravitational field runs at a slower rate.
- 4 I can calculate the Schwarzschild radius of a mass.

Daily Review



- 1 State Newton's universal law of gravitation.
- 2 Write the relationship for gravitational potential.
- 3 State the definition of escape velocity.
- 4 Write the relationship for escape velocity.
- 5 What is the gravitational potential at an infinite distance?

PART 1

Why is gravity not a force?

Newton's model gives the wrong orbit for Mercury.

By the end of this part...



1

I can state that special relativity deals with inertial frames and general relativity with non-inertial frames.

2

I can state the equivalence principle word for word.

3

I can describe what the equivalence principle means for light and for clocks.

Special and general relativity



Special relativity

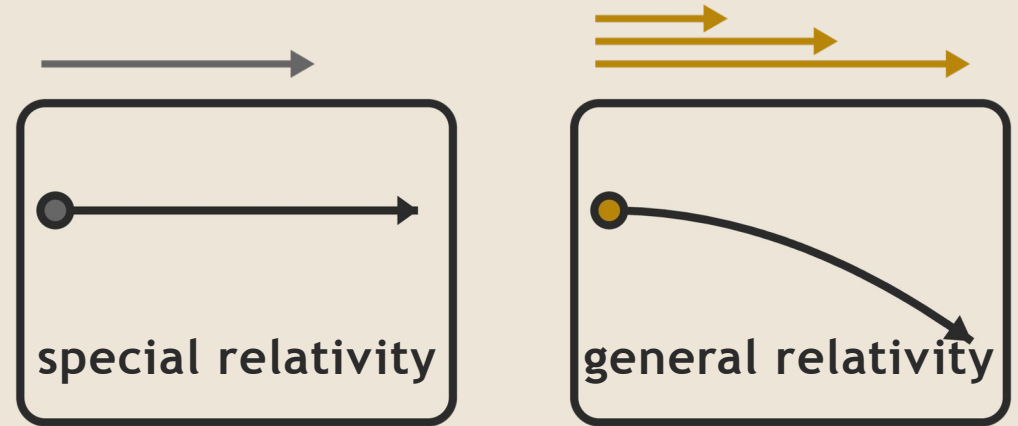
Inertial frames.

General relativity

Non-inertial frames —
accelerating.

The test

A ball thrown across bends
only if you accelerate.



Constant velocity left, uniform
acceleration right.

The equivalence principle



LEARN THIS WORD FOR WORD

It is not possible to distinguish between the **effects** on an **observer** of a uniform **gravitational field** and of a **constant acceleration**.

IOP SCOTLAND

The next slides are IOP Scotland's and Perimeter Institute's own, kept exactly as they made them. We stop after each section and write our own notes.

Taught in class from their slides

NOT IN THIS PDF

41 slides of this lesson are IOP Scotland's and Perimeter Institute's own work. They are shown in class and are not reproduced here.

The three JOTTER NOTES pages that follow are the notes we take from them, and they carry the whole of the equivalence principle.

Why gravity is not a force



Newton's force model gives Mercury the wrong orbit.

It also needs gravity to act instantly, faster than light.

Einstein: what we call gravity is acceleration, not a force.

Freefall in a uniform field is the same as floating in space.

Standing still in that field is the same as accelerating.

The effects of gravity and of acceleration cannot be told apart.

Why light bends



Light travels in a straight line through spacetime.

At constant speed it crosses to the same height on the far wall.

In an accelerating cabin the far wall rises as it crosses.

So the pulse lands lower, and the observer inside sees it bend.

By the equivalence principle, light bends in a field too.

Light is not pulled. We accelerate through curved spacetime.

Clocks in a gravitational field



Light pulses leave the back of the cabin at a steady rate.
As the cabin accelerates, the front pulls away from them.
The pulses arrive more slowly, so the back clock looks slow.
By the equivalence principle, the same happens in a field.
A clock lower in a gravitational field runs at a slower rate.
This is why a GPS clock in orbit runs fast.

Watch out — say rate, and justify it

THE MISTAKE

- "Time passes faster at the top of the hill."
- "The correction is $-39\ \mu\text{s}$."
- "The clock is slower because the probe is moving."

THE CORRECT ANSWER

- "Time passes at a faster rate at the top of the hill."
- "The clock is reduced by $39\ \mu\text{s}$ per day."
- "The field is weaker, so the clock runs at a faster rate."

Look back — by the end of this part...



- 1 I can state that special relativity deals with inertial frames and general relativity with non-inertial frames.
- 2 I can state the equivalence principle word for word.
- 3 I can describe what the equivalence principle means for light and for clocks.

Reading and homework



Scholar Topic 6

Scholar: Rotational motion, Topic 6.
BrightRed Study Guide pp. 34, 35, 38, 39.
Past paper: 2016 Q5.



Scholar

Daily Review



- 1 State the equivalence principle.
- 2 Which relativity deals with accelerating frames?
- 3 Why does a pulse of light bend inside an accelerating spacecraft?
- 4 Does a clock lower in a gravitational field run faster or slower?
- 5 Give one piece of evidence that Newton's force model of gravity fails.

PART 2

So what is gravity?

A bend in spacetime, and a straight line across it.

By the end of this part...



1

I can describe spacetime as three dimensions of space and one of time.

2

I can draw and read world lines for objects that are stationary, moving at constant velocity and accelerating.

3

I can explain what a geodesic is and why light follows one.

Spacetime



Three and one

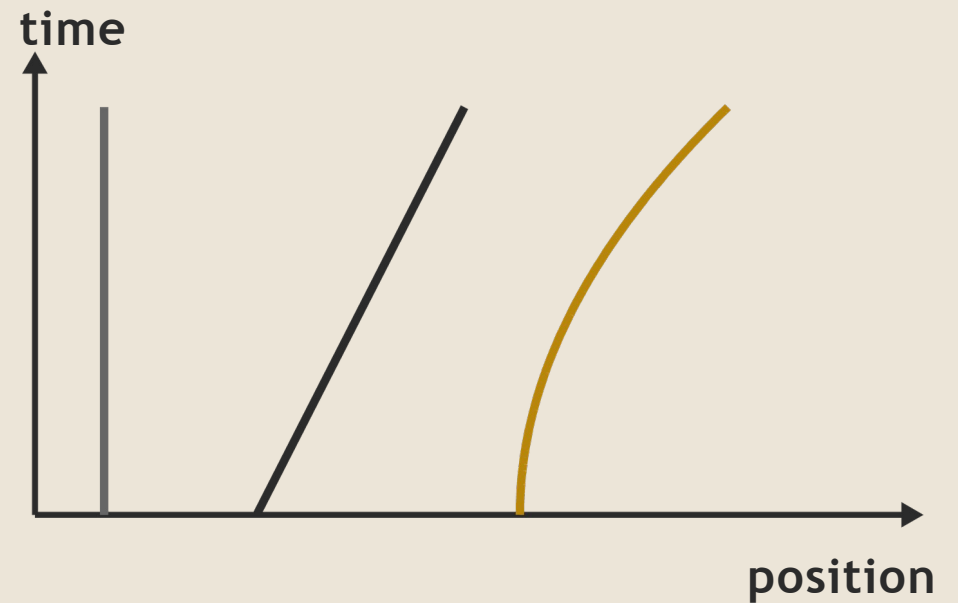
Three space dimensions, one of time.

On a diagram

Space across, time up.

A world line

An object's whole history, drawn as one line.



World lines



Stationary

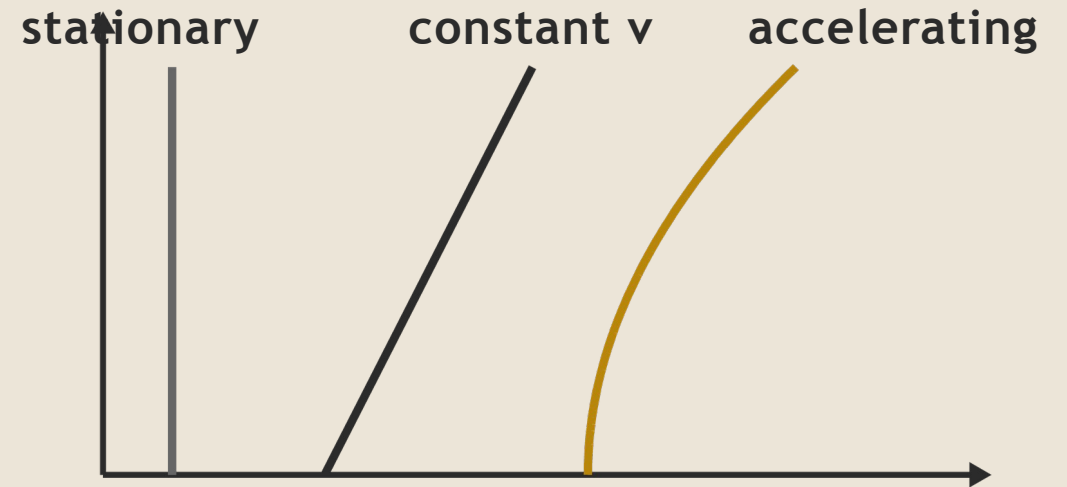
A vertical line.

Constant velocity

A straight, sloping line.

Accelerating

A curved line. The curve is what matters, not the lean.



Time up the page, position across it.

Reading world lines in the exam

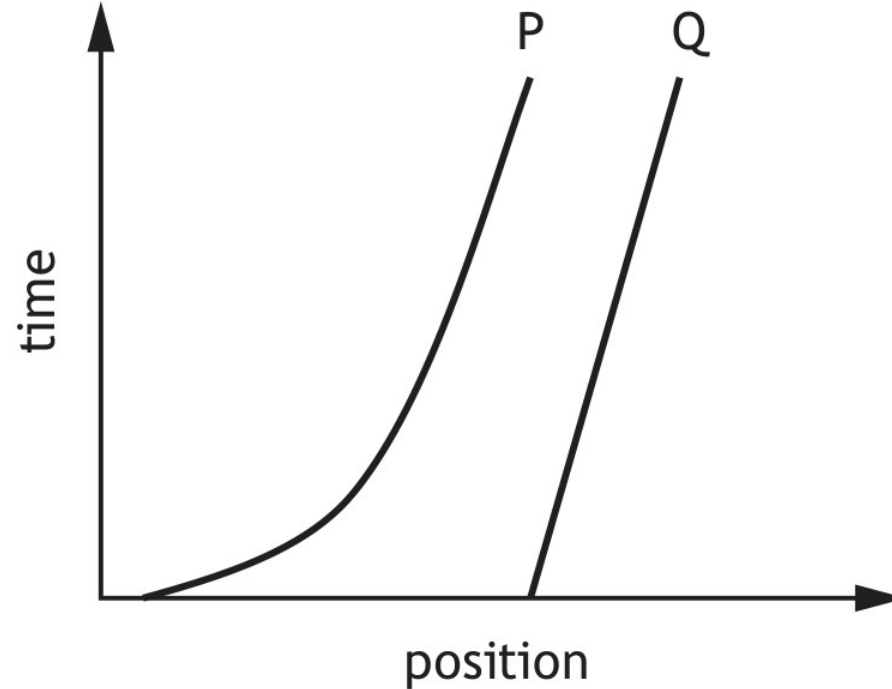


Figure 6A

2019 Q6(b): P is accelerating — its world line is curved.

Light cones



Light's world lines

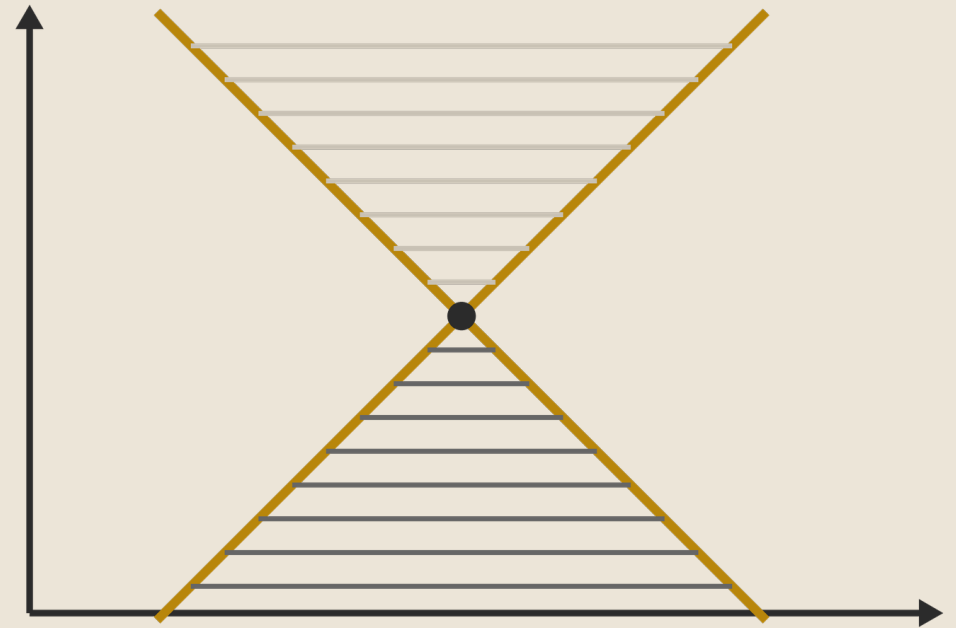
Always at a fixed angle.

The limit

Nothing with mass leans further over.

Past and future

All this event can see, or be seen by.



Mass curves spacetime



The idea

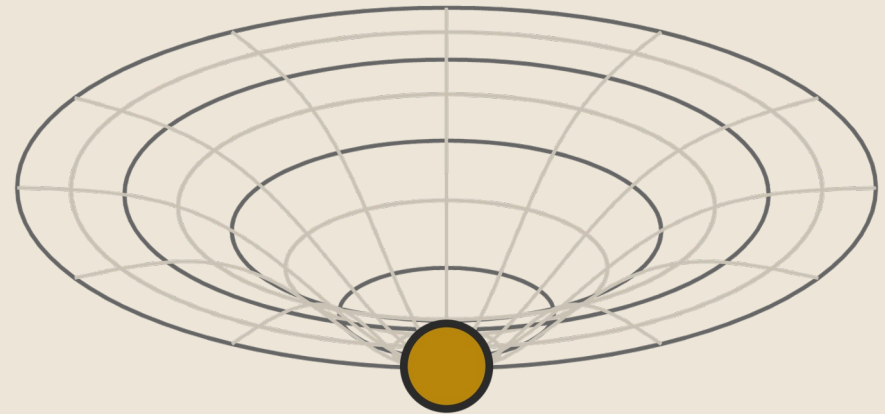
Mass curves spacetime;
gravity is that curvature.

Rubber sheet

A ball dents it; a marble
bends past.

Not a pull

The surface is bent.



Evidence – the orbit of Mercury

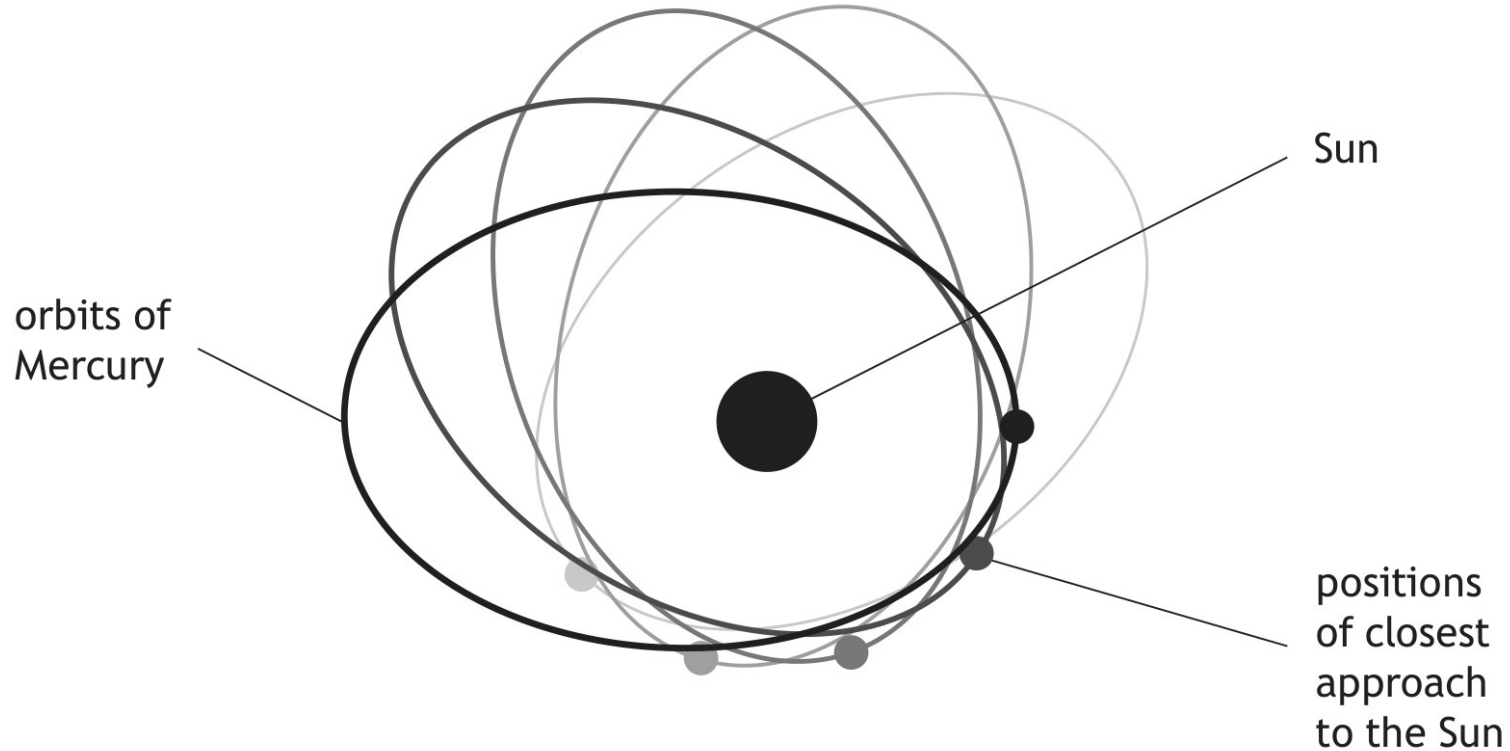


Figure 5A

Mercury's perihelion precesses – Newton's model gets it wrong.

Geodesics



Definition

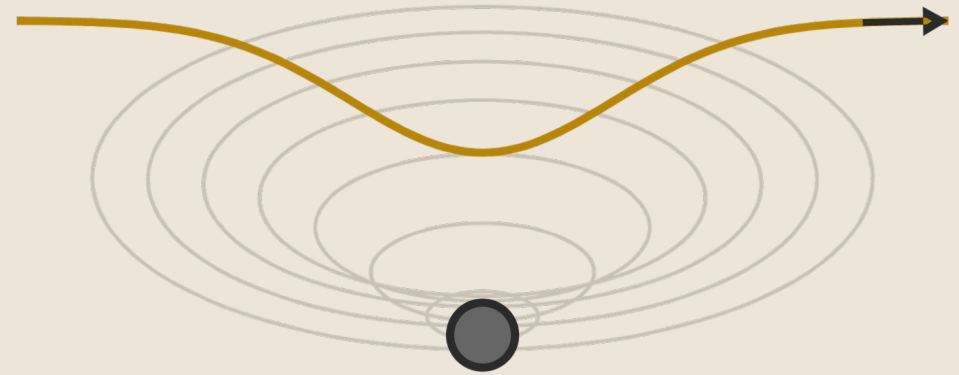
The shortest path between two points in spacetime.

Who follows one

Light, or anything moving freely.

Why it looks bent

Spacetime is curved.



Gravitational lensing



What happens

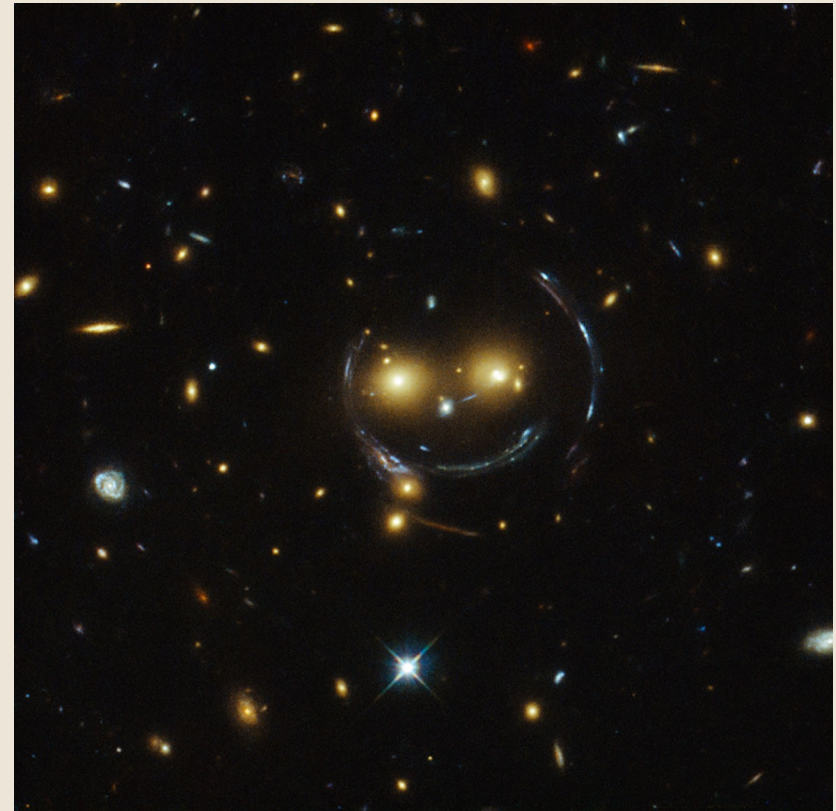
Distant light follows curved geodesics.

What you see

Arcs, or a ring, around the mass in front.

Why it matters

Mass curves spacetime.



SDSS J1038+4849, the "smiling" lens (Hubble).

Look back — by the end of this part...



1

I can describe spacetime as three dimensions of space and one of time.

2

I can draw and read world lines for objects that are stationary, moving at constant velocity and accelerating.

3

I can explain what a geodesic is and why light follows one.

Reading and homework



Scholar Topic 6

Scholar: Rotational motion, Topic 6.
BrightRed Study Guide pp. 36, 37, 40, 41.
Past paper: 2023 Q5, open-ended.



Scholar

Daily Review



- 1 What are the four dimensions of spacetime?
- 2 What does a curved world line tell you about an object?
- 3 State what a geodesic is.
- 4 Why does light follow a curved path near a massive object?
- 5 Name one piece of evidence that mass curves spacetime.

PART 3

What if nothing can escape?

Squeeze a mass small enough and even light cannot get out.

By the end of this part...



- 1 I can state that the escape velocity from the event horizon of a black hole is the speed of light.
- 2 I can state what the Schwarzschild radius of a black hole is.
- 3 I can calculate the Schwarzschild radius of a mass.

The event horizon



Escape velocity

The least velocity needed to escape to infinity.

At the horizon

It equals the speed of light.

So

Nothing inside can get out.



The Schwarzschild radius



Either form scores

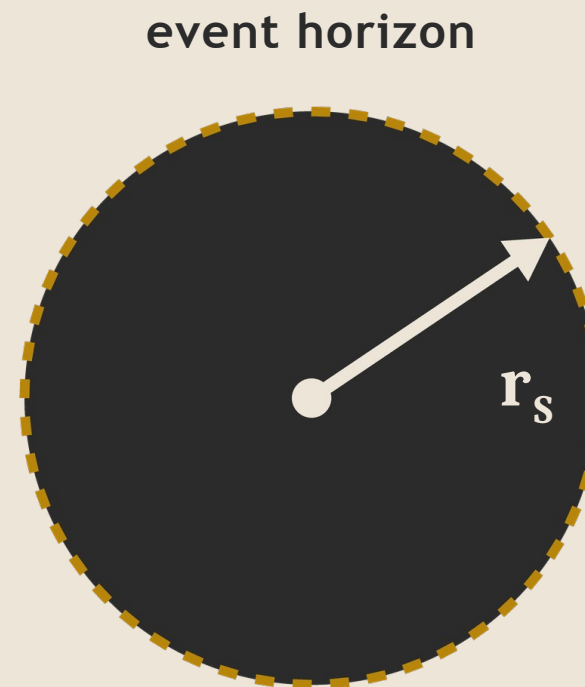
Centre of the mass to the event horizon.

Or

Where escape velocity would equal c .

Every mass has one

Not only black holes.



Calculating the Schwarzschild radius



$$r_s = 2GM/c^2$$

r_s

Schwarzschild
radius (m)

G

gravitational
constant

M

mass (kg)

c

speed of light
(m s⁻¹)

Both G and c are on the relationships sheet.

Worked example 1 – the Sun

The mass of the Sun is 2.0×10^{30} kg.

Calculate the Schwarzschild radius of the Sun.

$$M = 2.0 \times 10^{30} \text{ kg}$$

$$r_s = ?$$



$$r_s = 2GM/c^2$$

$$r_s = 2 \times 6.67 \times 10^{-11} \times 2.0 \times 10^{30} \\ \div (3.00 \times 10^8)^2$$

$$\underline{r_s = 3.0 \times 10^3 \text{ m}}$$

Worked example 2 – the Earth

The mass of the Earth is 6.0×10^{24} kg.

Calculate the Schwarzschild radius of the Earth.

$$M = 6.0 \times 10^{24} \text{ kg}$$

$$r_s = ?$$



$$r_s = 2GM/c^2$$

$$r_s = 2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \\ \div (3.00 \times 10^8)^2$$

$$\underline{r_s = 8.9 \times 10^{-3} \text{ m}}$$

Time at the event horizon



From far away

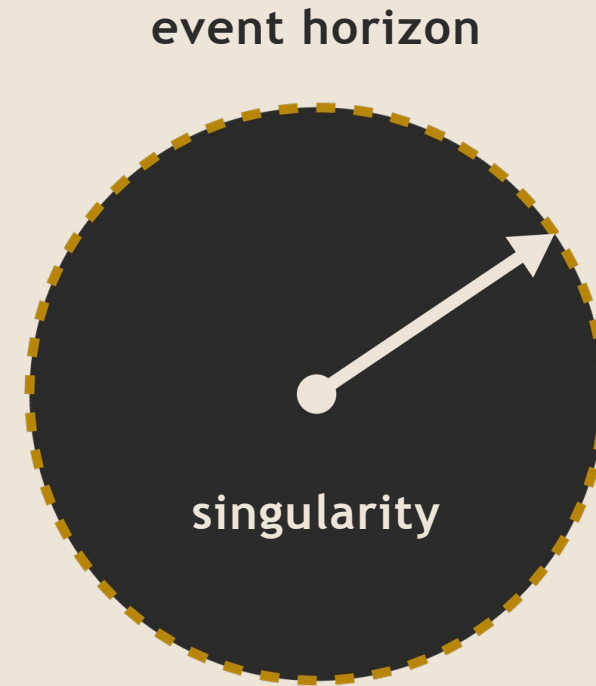
Time appears frozen at the horizon.

Why

Its light takes ever longer to climb out.

Falling in

Your own clock is normal.



M87* — a real event horizon



The dark centre is the event horizon.

GPS clocks



Two effects

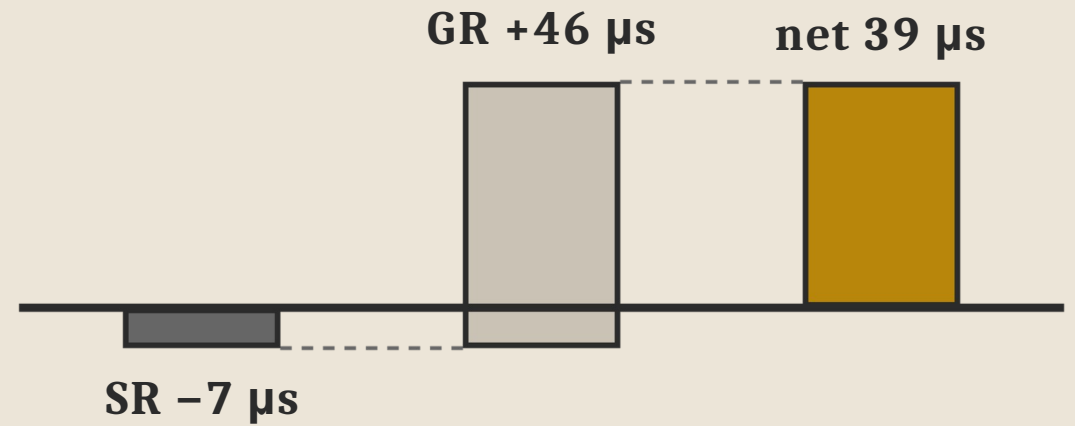
Speed slows the clock; the weaker field wins.

Net effect

It gains about $39\text{ }\mu\text{s}$ a day.

Why it matters

Light goes 12 km in $39\text{ }\mu\text{s}$.



The satellite clock, against one on the ground.

The GPS correction in the exam

Effect	Per day	Why
Special relativity	$-7\ \mu\text{s}$	The satellite is moving
General relativity	$+46\ \mu\text{s}$	The field is weaker in orbit
Net	$+39\ \mu\text{s}$	The clock gains, so it is reduced

Where you meet it



An artist's impression, not a photo.

Look back — by the end of this part...



- 1 I can state that the escape velocity from the event horizon of a black hole is the speed of light.
- 2 I can state what the Schwarzschild radius of a black hole is.
- 3 I can calculate the Schwarzschild radius of a mass.

Reading and homework



Scholar Topic 6

Scholar: Topic 6, then the Unit 1 test.
BrightRed Study Guide pp. 42, 43.
Past paper: 2022 Q5(a).



Scholar

Experiments you could try

- Stretch a sheet over a frame, drop a heavy ball on it and roll marbles past.
- Work out the daily drift of a GPS clock from the two effects, and what distance error it would cause.
- Compare a phone's clock with a radio time signal over a week.
- Look up Mercury's measured precession and compare it with Newton's prediction.

Check yourself

- 1 State the equivalence principle, word for word.
- 2 Sketch world lines for a stationary, a constant-velocity and an accelerating object.
- 3 Explain why light bends near a massive object.
- 4 Explain why a clock lower in a field runs at a slower rate.
- 5 Calculate the Schwarzschild radius of a 1.0×10^{31} kg star.

Course specification – 1 of 4

- Knowledge that special relativity deals with motion in inertial (non-accelerating) frames of reference and that general relativity deals with motion in non-inertial (accelerating) frames of reference.
- Statement of the equivalence principle (that it is not possible to distinguish between the effects on an observer of a uniform gravitational field and of a constant acceleration) and knowledge of its consequences.

Course specification – 2 of 4

- Consideration of spacetime as a unified representation of three dimensions of space and one dimension of time.
- Knowledge that general relativity leads to the interpretation that mass curves spacetime, and that gravity arises from the curvature of spacetime.
- Knowledge that light or a freely moving object follows a geodesic (the path with the shortest distance between two points) in spacetime.

Course specification — 3 of 4

- Representation of world lines for objects which are stationary, moving with constant velocity and accelerating.
- Knowledge that the escape velocity from the event horizon of a black hole is equal to the speed of light.
- Knowledge that, from the perspective of a distant observer, time appears to be frozen at the event horizon of a black hole.

Course specification — 4 of 4

- Knowledge that the Schwarzschild radius of a black hole is the distance from its centre (singularity) to its event horizon.
- Use of an appropriate relationship to solve problems relating to the Schwarzschild radius of a black hole.

Credits (1 of 3)

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The equivalence principle: © Qualifications Scotland (SQA)

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Watch out – say rate, and justify it: 2020 Q4(b), 2024 Q5(b) and 2025 Q4(c) marking instructions. © Qualifications Scotland (SQA)

Reading and homework: SCHOLAR AH Physics © Heriot-Watt University; BrightRed Advanced Higher Physics Study Guide

Reading world lines in the exam: 2019 Q6(b), Figure 6A. © Qualifications Scotland (SQA)

Evidence – the orbit of Mercury: 2018 Q5(c), Figure 5A. © Qualifications Scotland (SQA)

Gravitational lensing: Gravitational lens SDSS J1038+4849: NASA/ESA Hubble

Credits (2 of 3)

The Schwarzschild radius: 2018 Q5(a) marking instructions. © Qualifications Scotland (SQA)

Worked example 1 – the Sun: 2018 Q5(b)(i).
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Worked example 2 – the Earth: 2024 Q5(a)(i).
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M87* – a real event horizon: M87*: Event Horizon Telescope Collaboration, CC BY 4.0, via Wikimedia Commons

The GPS correction in the exam: 2024 Q5(b).
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Where you meet it: GPS Block IIF satellite:
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Credits (3 of 3)

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